Avi Vajpeyi

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Professional Experience

Postdoctoral Research Fellow (Statistics) — University of Auckland
Bayesian noise modeling and evidence estimation, with Prof. Renate Meyer. — 2026–present

Postdoctoral Research Fellow (Physics) — University of Auckland
Gravitational-wave data analysis for the LVK and LISA. — 2023–2026

Education

Ph.D. in Astrophysics (Advisors: Dr. R. Smith, Prof. E. Thrane) — Monash University
Thesis: "Astrophysical Inference in the Data Driven Era" — 2019–2022

B.A. in Computer Science and Physics (GPA: 4.0/4.0, Valedictorian) — The College of Wooster
Thesis: "Accelerating Granular Flow on GPUs with OpenCL & CUDA" — 2014–2018

Selected Research

Multivariate P-spline spectral estimation — arXiv preprint (2026)
Flexible spectral-matrix estimation for correlated multivariate time series. — arXiv:2607.04833

Differentiable WDM transform — arXiv preprint (2026)
Differentiable time-frequency methods for gravitational-wave and LISA analysis. — arXiv:2606.20269

Variational inference for correlated noise — Physical Review D (2025)
Scalable Bayesian inference for correlated gravitational-wave detector networks. — PRD 111, 062003

Bayesian P-spline modelling of LISA noise — Physical Review D (2026)
Non-parametric spectral-density estimation using penalised splines. — arXiv:2510.00533

Evidence estimation via density approximation — Physical Review D (2026)
Evidence estimation using informed approximations to posterior densities. — arXiv:2512.10283

For a complete publication record, see Google Scholar.

Research Software

- bilby and bilby_pipe
  Contributed maintenance to widely used Bayesian-inference and gravitational-wave analysis software.
- parallel_bilby — LVK parameter estimation — 2019–2023
  Co-developer of the LVK's massively parallel Bayesian inference software, used for its most computationally expensive analyses (GWTC catalogs); led its ADACS-funded upgrade for the fourth observing run.
- asimov — LVK analysis workflow framework — 2021–2023
  Contributor to the framework coordinating the LVK's parameter-estimation workflows.
- COMPAS — compact-object population synthesis
  Contributed maintenance and scientific-computing support.

Industry Experience

Software Consultant — Quaisr
Built data-agnostic ML/active-learning blocks linking TensorFlow backends to visual workflows. 2022–2023

Software Engineer — Bamboo Learning Inc
Developed NLP components and maths/listening-comprehension content for Alexa. Jul 2018–Mar 2019

Machine Learning Consultant — MOAD
Deployed CNN image classifiers and delivered hands-on machine-learning tutorials. Jul–Aug 2018

Software Engineering Practicum — Gitkeeper
Built and tested Python backend components for Git-based automated assessment. Aug 2016–Feb 2017

Publications

[1] N. Aimen, P. Maturana-Russel, **A. Vajpeyi**, N. Christensen, and R. Meyer, *Bayesian P-spline recovery of stochastic gravitational-wave backgrounds in LISA*, Aug. 21, 2026. arXiv: 2608.20629. [Online]. Available: https://arxiv.org/abs/2608.20629

[2] N. Aimen, P. Maturana-Russel, **A. Vajpeyi**, N. Christensen, and R. Meyer, "Bayesian power spectral density estimation for LISA noise based on penalized splines with a parametric boost", *Physical Review D*, vol. 113, no. 2, p. 024 022, Jan. 2026. arXiv: 2510.00533 [gr-qc]. [Online]. Available: https://doi.org/10.1103/dcb6-1jsl

[3] A. Boumerdassi, M. C. Edwards, **A. Vajpeyi**, and O. Burke, "First-time assessment of glitch-induced bias and uncertainty in inference of extreme mass ratio inspirals", *Physical Review D*, vol. 113, no. 10, p. 104 044, May 19, 2026. arXiv: 2512.16322 [gr-qc]. [Online]. Available: https://doi.org/10.1103/9678-764y

[4] IceCube Collaboration, LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, *Deep Search for Joint Sources of Gravitational Waves and High-Energy Neutrinos with IceCube During the Third Observing Run of LIGO and Virgo*, Jan. 12, 2026. arXiv: 2601.07595. [Online]. Available: https://arxiv.org/abs/2601.07595

[5] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "All-sky search for long-duration gravitational-wave transients in the first part of the fourth LIGO-Virgo-KAGRA observing run", *Physical Review D*, vol. 113, no. 8, p. 082 004, Apr. 27, 2026. arXiv: 2507.12282. [Online]. Available: https://doi.org/10.1103/83j3-pgk1

[6] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Constraints on Gravitational Waves from the 2024 Vela Pulsar Glitch", *The Astrophysical Journal*, vol. 1005, no. 1, p. 12, Jun. 19, 2026. arXiv: 2512.17990. [Online]. Available: https://doi.org/10.3847/1538-4357/ae63bc

[7] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Cosmological and High-Energy Physics Implications from Gravitational-Wave Background Searches in LIGO-Virgo-KAGRA's O1–O4a Runs", *Physical Review X*, vol. 16, no. 4, p. 041 023, Oct. 22, 2026. arXiv: 2510.26848. [Online]. Available: https://doi.org/10.1103/dlfv-z6m2

[8] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Direct Multimodel Dark Matter Search with Gravitational-Wave Interferometers Using Data from the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run", *Physical Review Letters*, vol. 137, no. 13, p. 131 001, Sep. 22, 2026. arXiv: 2510.27022. [Online]. Available: https://doi.org/10.1103/qdnd-rsnh

[9] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Directional search for persistent gravitational waves: Results from the first part of LIGO-Virgo-KAGRA's fourth observing run", *Physical Review D*, vol. 114, no. 2, p. 022 001, Jul. 6, 2026. arXiv: 2510.17487. [Online]. Available: https://doi.org/10.1103/wgp7-t8r8

[10] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "GW230814: investigation of a loud gravitational-wave signal observed with a single detector", *The Astrophysical Journal Letters*, vol. 1004, no. 2, p. L23, Jun. 2026. arXiv: 2509.07348. [Online]. Available: https://doi.org/10.3847/2041-8213/ae2ad3

[11] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "GWTC-4.0: Constraints on the Cosmic Expansion Rate and Modified Gravitational-wave Propagation", *The Astrophysical Journal Letters*, vol. 1007, no. 1, p. L17, Aug. 5, 2026. arXiv: 2509.04348. [Online]. Available: https://doi.org/10.3847/2041-8213/ae69dd

[12] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "GWTC-4.0: Methods for Identifying and Characterizing Gravitational-wave Transients", *The Astrophysical Journal Letters*, vol. 1004, no. 2, p. L21, Jun. 18, 2026. arXiv: 2508.18081. [Online]. Available: https://doi.org/10.3847/2041-8213/ae447b

[13] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "GWTC-4.0: Population Properties of Merging Compact Binaries", *The Astrophysical Journal Letters*, vol. 1005, no. 2, p. L51, Jul. 3, 2026. arXiv: 2508.18083. [Online]. Available: https://doi.org/10.3847/2041-8213/ae771e

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[18] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Open Data from LIGO, Virgo, and KAGRA through the First Part of the Fourth Observing Run", *The Astrophysical Journal*, vol. 1004, no. 2, p. 232, Jun. 18, 2026. arXiv: 2508.18079. [Online]. Available: https://doi.org/10.3847/1538-4357/ae211e

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[21] J. Liu, **A. Vajpeyi**, R. Meyer, J. E. Lee, and P. Maturana-Russel, *Variational Bayesian Inference for the Spectral Structure of LISA Noise*, Aug. 23, 2026. arXiv: 2608.22245. [Online]. Available: https://arxiv.org/abs/2608.22245

[22] Y. Liu, R. Meyer, N. Christensen, J. E. Lee, J. Liu, P. Maturana-Russel, and **A. Vajpeyi**, *Bayesian nonparametric estimation of correlated gravitational wave detector network noise using matrix-gamma process priors*, Jul. 29, 2026. arXiv: 2607.26619. [Online]. Available: https://arxiv.org/abs/2607.26619

[23] **A. Vajpeyi**, G. Mentasti, Q. Baghi, O. Burke, and L. Speri, *An explicit and differentiable Wilson-Daubechies-Meyer transform for gravitational-wave data analysis*, Jun. 18, 2026. arXiv: 2606.20269 [gr-qc]. [Online]. Available: https://arxiv.org/abs/2606.20269

[24] **A. Vajpeyi**, R. Meyer, P. Maturana-Russel, and J. Liu, *Multivariate Bayesian P-spline estimation of spectral density matrices, with application to LISA TDI noise*, Jul. 6, 2026. arXiv: 2607.04833 [stat.ME]. [Online]. Available: https://arxiv.org/abs/2607.04833

[25] E. M. Zahraoui, P. Maturana-Russel, **A. Vajpeyi**, W. van Straten, R. Meyer, and S. Gulyaev, "Enhancing evidence estimation through informed probability density approximation", *Physical Review D*, vol. 113, no. 8, p. 083 014, Apr. 10, 2026, Also known as "MorphZ". arXiv: 2512.10283 [stat.ME]. [Online]. Available: https://doi.org/10.1103/1554-y6ns

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[34] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Search for Continuous Gravitational Waves from Known Pulsars in the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run", *Astrophys. J.*, vol. 983, no. 2, p. 99, 2025, Art. no. 99.

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[37] J. Liu, **A. Vajpeyi**, R. Meyer, K. Janssens, J. E. Lee, P. Maturana-Russel, N. Christensen, and Y. Liu, "Variational inference for correlated gravitational wave detector network noise", *Phys. Rev. D*, vol. 111, no. 6, p. 062 003, Mar. 2025, Art. no. 062003.

[38] Team COMPAS, I. Mandel, J. Riley, A. Boesky, A. Brcek, R. Hirai, V. Kapil, M. Y. M. Lau, J. D. Merritt, N. Rodriguez-Segovia, I. Romero-Shaw, Y. Song, S. Stevenson, **A. Vajpeyi**, L. A. C. van Son, A. Vigna-Gomez, and R. Willcox, "Rapid stellar and binary population synthesis with COMPAS: Methods paper II", *Astrophys. J. Suppl.*, vol. 280, no. 1, p. 43, 2025, Art. no. 43. arXiv: 2506.02316 [astro-ph.SR].

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[43] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "GWTC-3: Compact Binary Coalescences Observed by LIGO and Virgo during the Second Part of the Third Observing Run", *Phys. Rev. X*, vol. 13, p. 041 039, 2023, Art. no. 041039.

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[47] **A. Vajpeyi**, R. Smith, and E. Thrane, "Deep Follow-up for Gravitational-wave Inference: A Case Study with GW151226", *Astrophys. J.*, vol. 947, no. 1, p. 10, Apr. 2023, Art. no. 10. arXiv: 2203.13406 [gr-qc].

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Talks

- **International Training Workshop on Frontiers from Quanta to Cosmos Physics (FQCP2026), ICTP-AP / UCAS, Beijing** Aug–Sep 2026
  Delivered a training lecture for Master's and PhD students on Bayesian parameter estimation for compact binary coalescences in LIGO and LISA.
- **2nd New Horizons for Psi Workshop, Beijing** 25 Aug 2026
  Accelerated Bayesian noise estimation for multivariate and non-stationary LISA data.
- **AUT Mathematical Modelling and Analytics Symposium** Nov 2025
  Multivariate time-series spectral analysis with splines; applications to EEG, financial, and astrophysical data.
- **BayesComp25 Invited Session** 2025
  Bayesian computation in astrophysics
- **COMPAS** Nov 2024
  Active Machine Learning with Deep GPs for COMPAS Population Inference
- **Flatiron Institute** Jul 2024
  Mind the Gaps — Time-Frequency Approaches for analysing LISA data with gaps
- **NZ Gravity Workshop, Canterbury** Aug 2023
  Tutorial on parallelisation techniques
- **NZ Summer School on Gravitational Waves** Feb 2023
  Bayesian parameter estimation for compact-binary inspiral signals
- **LIGO Rates and Populations** Oct 2021
  Measuring properties of AGN disks from BBH GW
- **FIT3170 Guest Lecture** Apr 2021
  UI/UX Design for programmers
- **OzGrav Online** Jun 2020
  Understanding our universe with parallel computing
- **OzGrav Retreat** Nov 2019
  A High Mass BH Search
- **Mt B. Observatory** Feb 2019
  GPU Computing in Astro
- **Google I/O Extended Bronx** Jul 2018
  "Icebreakers" tech demo

Outreach

- Assist UoA, Monash, OzGrav with public demos (open days, outreach, talks).
- Coded games Moonshot, three-body, looper, polarity, for outreach.

Public Talks

- Auckland Astronomical Society Aug 2025
  "Celestial Songs: A Cosmic Spacetime Symphony" — guest lecture, Stardome Observatory; recording.
- MOTAT STEM Fair — public engagement event 2025

- Raising the Bar (Auckland) Aug 2024

  "Celestial Songs: A Cosmic Spacetime Symphony" — Auckland Fish Market; traced cosmic ripples from Einstein's 1916 theories to modern gravitational-wave discoveries and New Zealand's role. Event / podcast.
- Future Me (Auckland) Jun 2024

  "Thump in the night — Statistics and the hunt for black holes".
- Pint of Science (Melbourne) 2019–2022

  "Black Holes: The Ultimate Snitches" — how gravitational waves from colliding black holes let us study the dust-veiled centres of galaxies; delivered while a PhD candidate at Monash University.

Media

- 95bFM — Ready Steady Learn, radio interview Aug 2024
- RNZ — The science of space sounds, radio interview Aug 2024

Research Funding and Proposals

- **Dumont d'Urville NZ–France Science & Technology Support Programme** 2026

  Principal Investigator; application pending.
- **ADACS Merit Allocation Program** (2 successful awards, ≈AUD $20K in-kind) 2021, 2023

  One semester of professional software-development support per project: parallel_bilby and tess-atlas.

Hackathons

- Completed projects in 20 hackathons; published 19 games totaling +60Hrs of play.
- itch.io 1-BIT JAM #4 Oct 2024

  Towerfall: 1st in Gameplay, 4th of 331 overall
- Facebook Community Challenge ($2000) Oct 2020

  Best Oceania Region Award
- Google I/O Extended Bronx Hack: 1st Place ("Google Action") Jul 2018
- HackOH5: 1st Place ($1000) Apr 2017

  TensorFlow embeddings of newspaper text (1856-2010) demonstrating semantic change.

Teaching

Monash University: Teaching Assistant and Course Support (postgraduate and undergraduate)

- [FIT:3170] Software engineering practice — *team mentoring (independent); one guest lecture* 2021–2022

  Independently mentored two agile teams of approximately 20 students each year. Monitored their use of the Scaled Agile Framework (SAFe), including documentation and meeting practices; advised on technical stacks; graded project work; recruited guest lecturers and external clients. Delivered a guest lecture on UI/UX design for programmers (Apr 2021).
  2021 Monash teaching report: student satisfaction 88/100 raw and 98/100 weighted, both in the "Very High" band (SETU: exceeding expectations), against a Teaching Faculty average of 82/100. The survey went to the full 97-student cohort, of whom I taught roughly 40; 16 of the 17 respondents rated the teaching High or Very High. Report available on request.

- [FIT:9136] Algorithms and programming foundations in Python (postgraduate) — *tutorials (independent delivery); assessment co-authoring* 2020

  Sole tutor in the room for hybrid and online tutorials on Python and algorithms; co-authored assignments and examinations with the teaching team for the course lecturer's review; built automated grading and graded assessments. Tutorial content rotated weekly among three teaching assistants.

- [FIT:1048] Fundamentals of C++ — *laboratory tutorials (independent delivery)* Jul–Nov 2019

  Ran laboratory sessions in C++ programming and walked students through code solutions.

Workshops and Short Courses — *designed and delivered independently*

- International Training Workshop on Frontiers from Quanta to Cosmos Physics, Beijing — *invited lecture (postgraduate)* Aug–Sep 2026

  Bayesian parameter estimation for compact binary coalescences in LIGO and LISA; delivered as part of FQCP2026 at ICTP-AP / UCAS.

- 2023 NZ Summer School on Gravitational Waves — *lecture and tutorial* Feb 2023

  Sole designer and presenter of instruction on Bayesian parameter estimation for inspiral signals, to advanced undergraduate and postgraduate students.

- NZ Gravity Workshop, University of Canterbury — *tutorial* Aug 2023

  Sole designer and presenter of a one-hour tutorial on parallelisation techniques, including multiprocessing and GPU programming.

The College of Wooster: Undergraduate Teaching Assistant — *laboratory and discussion sessions, delivered independently under a course instructor*

- Six courses across Physics, Computer Science, and Interdisciplinary Studies 2015–2018

  [PHYS:201] Modern Physics Lab, [PHYS:112] Calculus Physics 2 Lab, [CS:200] Algorithm Analysis, [CS:120] Data Structures and Algorithms, [IDPT:405] Entrepreneurship, [IDPT:1991] Global Engagement. Ran laboratory and discussion sessions, guided students through experiments, code, and data analysis, and supported scientific writing and presentation. Also privately tutored two advanced-placement high-school students in physics and calculus (Oct–Dec 2018).

Awards

- RTP Stipend (AUD $28K) 2019–2022
- Monash Tuition Scholarship (AUD $49K) 2019–2022
- Wooster Commencement Speaker Jun 2018
- The Jonas O. Notestein Prize — highest GPA in the graduating class
- The Arthur H. Compton Prize in Physics — highest standing in physics
- The Endowed Faculty Scholarship
- NCAA Honor Roll Dec 2017
- MCURCSM: Best Paper Presentation Oct 2017
- Dean's List and Scholarship 2014–2018
- Honour Societies: ΦBK, ΠME Jun 2017

References

Daniel Foreman–Mackey, Ph.D.
Research Engineer, Anthropic
dfm.io

Barry McKernan, Ph.D.
Professor, American Museum of Natural History
bmckernan@amnh.org

Rory Smith, Ph.D.
Research Fellow, Monash University
rory.smith@monash.edu

Student Supervision

Postgraduate — day-to-day research supervision alongside the students' official supervisors; not yet a formal co-supervisor appointment.

- Jianan Liu — PhD 2023–present
  University of Auckland; fast Bayesian methods (variational inference) for GW detector noise.
- Nazeela Aimen — PhD 2023–present
  University of Auckland; non-parametric Bayesian power spectral density estimation for LISA noise.
- Yukun Yang — MSc 2024
  University of Auckland; "Learning from Black Holes with Neural-net Surrogates".
- Jiaxuan (Steven) Zhou — MSc 2023
  Monash University; fast eccentric waveforms via reduced-order models and neural networks.
- Eyleen Amanda Sánchez García — MSc 2025
  Pontificia Universidad Católica de Valparaíso; CCSNe detection with normalising flows.
- Manfred Linton — Science Scholars 2023
  University of Auckland.

Undergraduate Capstone Team Mentoring (Monash FIT3170)

- OwlEyes 2021
  Final-year project team mentoring; Visual Testing for Android Apps.
- Automated grading tool 2021
  Final-year project team mentoring; FIT3170-project4-2021.
- MLVet 2022
  Final-year project team mentoring; video-editing app; MLVETDevelopers.
- RABIT 2022
  Final-year project team mentoring; FIT3170-FY-Project-7.